

Proof. 1. Standing hypotheses and the degenerate case. We assume $X \geq 0$ a.s. and $a > 0$. If $\mathbb{P}(X = 0) = 1$ then $\mathbb{E}[X] = 0$ and $\mathbb{P}(X \geq a) = 0$; thus (M) holds with equality. Henceforth we suppose $\mathbb{P}(X > 0) > 0$.

2. Indicator of the event $\{X \geq a\}$. Define the indicator

$$I(\omega) := \mathbf{1}_{\{X \geq a\}}(\omega) = \begin{cases} 1, & X(\omega) \geq a, \\ 0, & X(\omega) < a. \end{cases}$$

By definition of expectation of an indicator,

$$\mathbb{E}[I] = \mathbb{P}(X \geq a). \quad (2.1)$$

3. Pointwise bound. For every $\omega \in \Omega$,

$$I(\omega) \leq \frac{X(\omega)}{a}. \quad (3.1)$$

Indeed, if $X(\omega) \geq a$ then $I(\omega) = 1 \leq X(\omega)/a$; if $X(\omega) < a$ then $I(\omega) = 0 \leq X(\omega)/a$.

4. Taking expectations. Since both sides of (3.1) are non-negative random variables, monotonicity of expectation yields

$$\mathbb{E}[I] \leq \mathbb{E}\left[\frac{X}{a}\right]. \quad (4.1)$$

Linearity of expectation gives

$$\mathbb{E}\left[\frac{X}{a}\right] = \frac{1}{a} \mathbb{E}[X]. \quad (4.2)$$

Combining (4.1) and (4.2) we obtain

$$\mathbb{E}[I] \leq \frac{\mathbb{E}[X]}{a}. \quad (4.3)$$

5. Replacing $\mathbb{E}[I]$ by the probability. Using (2.1) in (4.3) we get

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a},$$

which is precisely Markov's inequality (M).

6. Equality case. Equality in (M) can occur only if equality holds in every step above. Equality in (4.1) requires equality in the pointwise bound (3.1) on a set of full probability, because for non-negative random variables $Y \leq Z$ a.s.

$$\mathbb{E}[Y] = \mathbb{E}[Z] \implies Y = Z \text{ a.s.} \quad (6.1)$$

Applying (6.1) with $Y = I$ and $Z = X/a$ yields

$$\frac{X}{a} = I \text{ a.s.}$$

Hence, for $\mathbb{P}$ -a.e. ω ,

$$X(\omega) = a I(\omega) = \begin{cases} a, & X(\omega) \geq a, \\ 0, & X(\omega) < a. \end{cases}$$

Thus X takes only the two values 0 and a ; equivalently, X is a Bernoulli-type random variable with

$$\mathbb{P}(X = a) = \mathbb{P}(X \geq a), \quad \mathbb{P}(X = 0) = 1 - \mathbb{P}(X \geq a).$$

In this situation $\mathbb{E}[X] = a \mathbb{P}(X \geq a)$, so the inequality becomes an equality.

7. Additional remarks. If $\mathbb{E}[X] = \infty$ the right-hand side of (M) is infinite and the inequality is trivial. The requirement $a > 0$ is essential; for $a \leq 0$ the statement either reduces to $\mathbb{P}(X \geq a) = 1$ or becomes undefined.

Therefore Markov's inequality holds for every non-negative random variable, with equality exactly when X assumes only the values 0 and a a.s. $\square$ $\square$